8.4 Recursion

Recursion is an algorithm design technique, closely related to induction. It is similar to iteration, but more powerful. Using recursion, we can solve a problem with a given input, by solving the instances of the problem with a part of the input.

Example 8.10. Customers are waiting in a line at a counter. The man at the counter wants to know how many customers are waiting in the line.

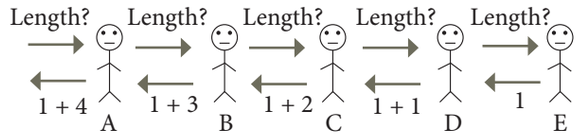


Figure 8.4: Length of a line

Instead of counting the length himself, he asks customer A for the length of the line with him at the head, customer A asks customer B for the length of the line with customer B at the head, and so on. When the query reaches the last customer in the line, E, since there is no one behind him, he replies 1 to D who asked him. D replies 1+1 = 2 to C, C replies 1+2 = 3 to B, B replies 1+3 = 4 to A, and A replies 1+4= 5 to the man in the counter

8.4.1 Recursive process

Example 8.10 illustrates a recursive process. Let us represent the sequence of 5 customers A, B, C, D and E as

[A,B,C,D,E]

The problem is to calculate the length of the sequence [A,B,C,D,E]. Let us name our solver length. If we pass a sequence as input, the solver length should output the length of the sequence.

length [A,B,C,D,E] = 5

Solver length breaks the sequence [A,B,C,D,E] into its first customer and the rest of the sequence.

first [A ,B,C,D,E] = A

rest [A ,B,C,D,E] = [B ,C,D,E]

To solve a problem recursively, solver length passes the reduced sequence [B,C,D,E] as input to a sub-solver, which is another instance of length. The solver assumes that the sub-solver outputs the length of [B,C,D,E], adds 1, and outputs it as the length of [A,B,C,D,E].

length [A,B,C,D,E] = 1 + length [B,C,D,E]

Each solver

1. receives an input,
2. passes an input of reduced size to a sub-solver,
3. receives the solution to the reduced input from the sub-solver, and produces the solution for the given input

as illustrated in Figure 8.5.

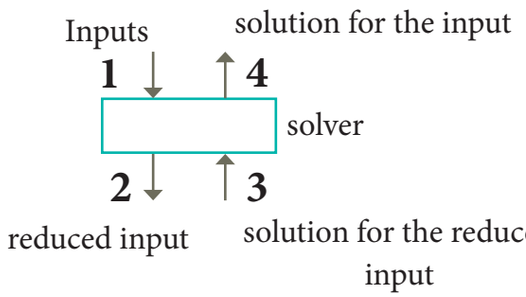


Figure 8.5: One instance of a solver in a recursive process

Figure 8.6 shows the input received and the solution produced by each

solver for Example 8.10. Each solver reduces the size of the input by one and passes it on to a sub-solver, resulting in 5 solvers. This continues until the input received by a solver is small enough to output the solution directly. The last solver received [E] as the input. Since [E] is small enough, the solver outputs the

Chapter 8 Page 102-114.indd 108 3/24/2020 9:15:11 AM